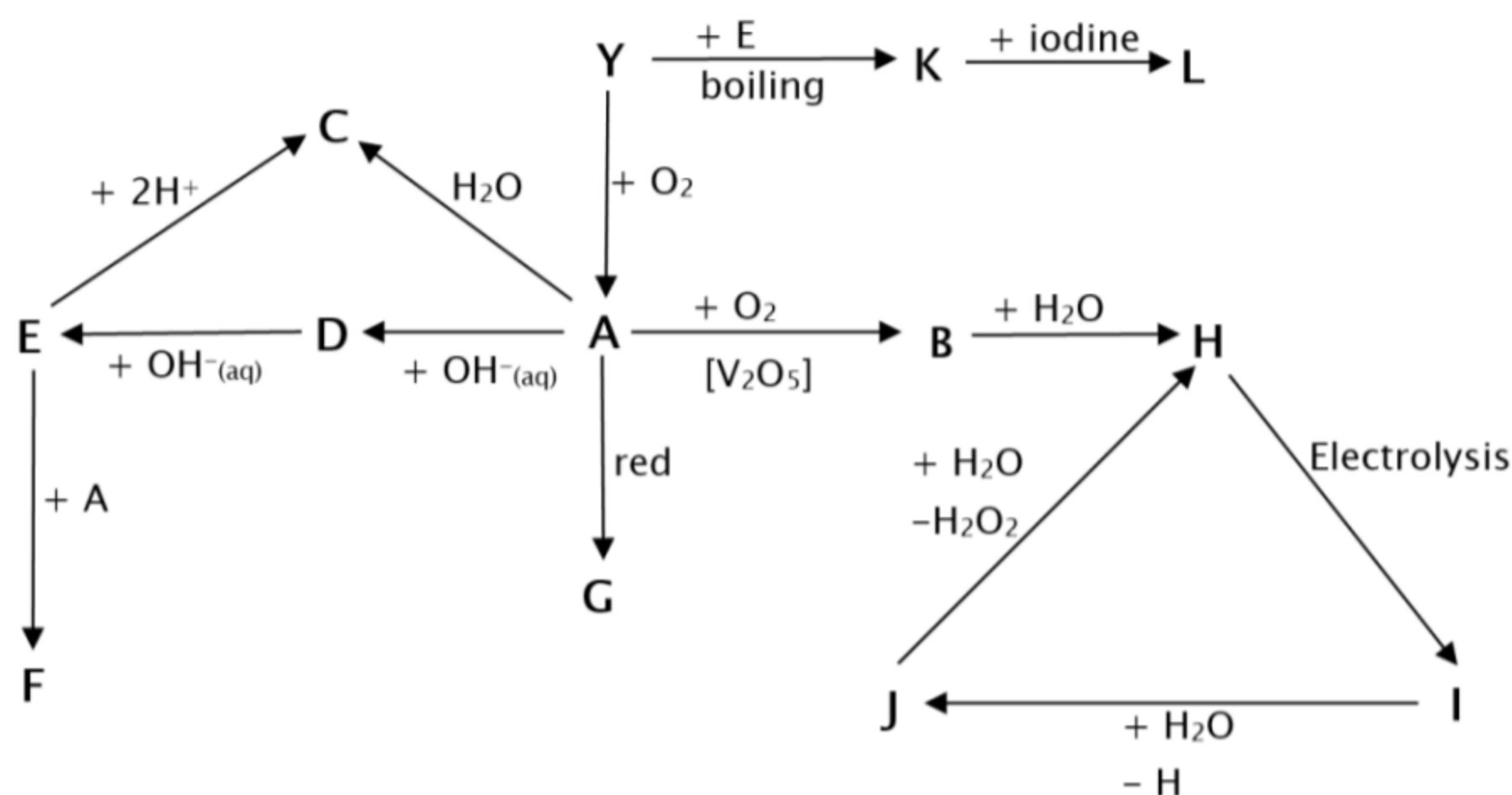


Nonmetal chemistry on connection with an allotropic element



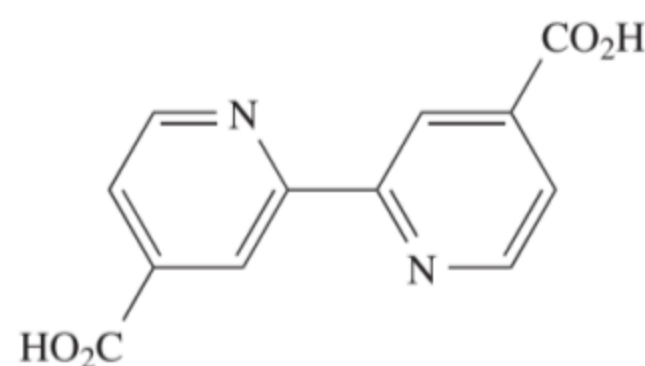
1. The molecules and particles **A** to **L** in the above scheme contain, aside from **Y**, only oxygen and/or hydrogen. Compound **B** is a technically extremely important substance, which condenses to give an ice-like solid below 17°C. It is the basis for the synthesis of the strong acid **H**. Compound **B** cannot be generated directly by combustion of element **Y**, but only by catalytic oxidation of compound **A**, a pungent smelling gas. This gas produces a moderately strong acid after reaction with water which cannot be isolated in water free form. If gas **A** reacts with magnesium hydroxide, a solution containing ions **D** will be generated. Using this solution, the company AG produces cellulose by boiling. By attachment of the Lewis-acid **A** to the Lewis-base **E** the ion **F** is produced. The potassium salt of **F** is used for disinfection, as in combination with water, gas **A** is generated.

Compound **A** may react with strong reducing agents to give acid **G**. The anion of acid **G** has strong reducing capability in alkaline solution, and thus instantaneously precipitates silver from a silver nitrate solution.

In the above-mentioned electrolysis, the anion of compound **I** is generated, which has the molecular formula $\text{H}_2\text{Y}_2\text{O}_8$. Hydrolysis of compound **I** delivers, after splitting off acid **H**, the so-called Caro's acid **J**. The anions **K** form a well-known silver complex.

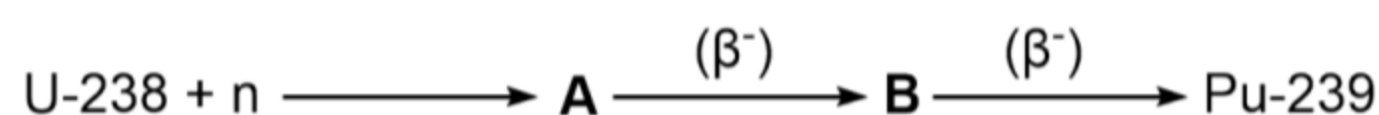
- 1.1. What is the element **Y**?
- 1.2. Write the formulae and the names of substances **A** to **L**.
- 1.3. Which particle is oxidized by electrolysis from **H** to **I**?
- 1.4. Give the name of the structure of ion **E** according to the VSEPR-theory and the bonding angle **E**.

2. A crucial component of a dye-sensitized solar cell (DSC) is the sensitizer. The latter captures photons which are converted to electric current in the cell. A typical dye is cis-[Ru(L)₂(NCS-N)₂] where L is the bpy derivative shown below:



- 2.1 What conformational change does the bpy ligand undergo when it binds to Ru²⁺?
- 2.2 An ideal dye should absorb light over the whole visible range, and the extinction coefficient should be high over the whole absorption range. Explain why this is.
- 2.3 The electronic transition associated with the absorption of cis-[Ru(L)₂(NCS-N)₂] in the visible region involves an electron in the metal t_{2g} level and a low-lying π* MO. What is the origin of the acceptor orbital, and what is the general name for this type of transition? Relate the type of transition to the redox properties of ruthenium(II).

3 Nuclear fuel is material used in nuclear power stations to produce heat to power turbines. Most nuclear fuels contain heavy fissile actinide elements that can undergo nuclear fission and sustain the chain reaction. The three most relevant fissile isotopes are uranium-233, uranium-235 and plutonium-239. The plutonium isotope 239 is inevitably produced in nuclear power plants operating with ²³⁸U-rich isotope mixtures, due to neutron capture.



- 3.1 Find the isotopes A and B.
- 3.2 State what naturally occurring element must be added to a nuclear reactor to produce U-233. Write out all nuclear reactions and decays that are part of U-233 production.

Geochronology is the science of determining the age of rocks, fossils, and sediments using signatures in the rocks themselves. Absolute geochronology can be accomplished through radioactive isotopes such as the Rb-Sr and the Sm-Nd system. In such “chronometer” systems the amount of daughter nuclide as a function of the amount of mother nuclide and time is given by:

$$N_D = N_M \cdot (e^{\lambda \cdot t} - 1) \quad (1)$$

and can be derived from the law of radioactive decay.

3.3 Derive equation (1) from above.

An interesting chronometer system is based on the decay of K-40. In addition, K-40 (natural abundance 0.012%) has a favorable half-life (1.25×10^9 years) for radiometric dating, which corresponds to 1/4 of the age of the earth. K-40 has a nuclide with an odd proton and neutron number, which leads to special decay properties. K-40 decays by β -decay (89.14%, $\lambda_b = 4.962 \times 10^{-10} \cdot \text{a}^{-1}$ (=per year)) to Ca-40 (decay energy 1.32 MeV) and by electron capture (0.2%) to Ar-40 (ground state, decay energy 1.51 MeV). Furthermore, the electron capture yields Ar-40 in an excited state (10.66%, decay energy 0.05 MeV). By γ -emission (γ -energy 1.461 MeV) the excited state of Ar-40 relaxes to the ground state. The decay constant for the overall electron capture process has been determined as $\lambda_e = 0.581 \times 10^{-10} \cdot \text{a}^{-1}$.

3.4 Draw the generic decay scheme of K-40.

The isotopic composition of Ar in rocks is not constant, but a function of their K content, age and thermal history. Provided that at time $t = 0$ (magmatic or metamorphic age), the sample contained no Ar-40 and from $t = 0$ to the present, the sample was a closed system for K and Ar, the ratio of K-40/Ar-40 can indicate the age of the rock. The analysis of 1 kg volcanic rock showed a total potassium content of 0.14 kg. The amount of Ar-40 has been determined as 7.638×10^{-4} mg.

3.5 Based on the generic decay scheme of K-40, estimate the age of this sample.

4. Recently, Khoja Nasreddin decided to become a chemist. He began his journey in chemistry by studying the thermal decomposition of an unknown red salt **A**, even though such work was quite challenging for beginners.

Khoja Nasreddin carefully decomposed 1.00 g of **A** at -70°C (**Reaction 1**) in a closed 1.00 L reactor initially filled with argon at a pressure of 100.0 kPa. After the decomposition, the pressure inside the reactor increased to 106.42 kPa, and the solid residue consisted of two compounds, one of which was identified as salt **B**.

Following the first stage of decomposition, he gradually heated the reactor to 200°C . During heating, the solid residue first turned into a mixture of solid and liquid, then into a solely liquid phase, and finally, at 128°C , a solid phase reappeared. At this temperature, the pressure in the reactor was 260.63 kPa.

Upon reaching 200°C , **B** began to decompose (**Reaction 2**), and after complete decomposition, no solid residue remained in the reactor. The pressure at 200°C was 396.82 kPa. When he introduced a glowing splint into the reactor, it ignited vigorously.

Nasreddin then attempted to synthesise salt **C**, which differs from **A** only by its cation. The mass fraction of the metal in **C** is equal to 12.63%. However, according to various sources, **C** is difficult to obtain. Instead, he successfully synthesised salt **D** (containing a complex cation, the same metal as in **C**, and 9.83%

hydrogen by mass) by treating compound **E** (mass fraction of one of the elements in **E** is 4.21%) with gas **F** in liquid ammonia. He noted that at very low temperatures, **F** exists as a dark blue liquid.

To determine the amount of gas **F** required for the synthesis of 1.000 g of **D**, he passed as much **F** as necessary through 100 mL of potassium iodide solution (**Reaction 3**). For titration of a 10.00 mL aliquot of the resulting brown solution, 16.30 mL of sodium thiosulfate solution (**Reaction 4**) with a concentration of 0.0998 M was used. He also noted that when he was passing gas **F** through the iodide solution, another colourless gas was produced in the same amount as gas **F**.

4.1 Identify substances A–F.

4.2 Write the balanced equations of Reactions 1–4.

